双语全文翻译｜what is spatial history pub 020110

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第 1 页｜English original

spatial history lab1 February 2010
What is Spatial History?
Richard White1
The Spatial History Project at Stanford is part of a larger spatial turn in history. It is a humble and modest –and expensive-- attempt to do history in a different way. I want to emphasize modest and humble. Most so-called "turns" in history emphasize their revolutionary intent. I think that what we are doing is different, but we are not announcing the end of history as you know it or the extinction of the text or the narrative. Historians will continue to write books. Historians will continue to tell stories.

The Spatial History Project, however, does operate outside normal historical practice in five ways. First, our projects are collaborative. Many of the things that visitors to our web site see involve collaborations between an historian, graduate students and undergraduates, geographers, GIS and visualization specialists, data base architects, and computer scientists. The scholars involved in the Spatial History Project can write books by themselves, but they cannot do a spatial history project on the scale they desire alone: we lack the knowledge, the craft, and ultimately the time. Second, while many of our presentations involve language and texts, our main focus is on visualizations, and by visualizations I mean something more than maps, charts, or pictures. Third, these visualizations overwhelmingly depend on digital history. By digital

1Director, Stanford University Spatial History Project
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history all I mean is the use of computers. Digital history allows the exploitation of kinds of evidence and data bases that would be too opaque or too unwieldy to use without computers. It is all the stuff that we cannot narrate, or at least narrate without losing our audience. All historians run across such evidence in the archive. We look at them and toss them aside. For me railroad freight rate tables are the quintessential example. Fourth, these projects are open-ended: everything –both tools and data—becomes part of a scholarly commons to be added to, subtracted from, reworked and recombined. The final, and most critical aspect of our departure from professional norms is our conceptual focus on space. Historians by definition focus on time. Chronology will always remain at the heart of a discipline that seeks to explain change over time, but this has left historians open to the charge from geographers that they write history as if it took place on the head of a pin. The charge is not true, but sometimes it is uncomfortably close to being true. Many of the classic histories of the last half-century and more – from Fernand Braudel's Mediterranean World to William Cronon's Nature's Metropolis – have been spatial in the sense that changing spatial relations that best explain the pattern of changes over time. When William Cronon republishes the old

Figure 1 | Rates of Travel from New York City, 1830 and 1857 by Charles Puallin, 1932. From William Cronon's book, Nature's Metropolis: Chicago and the Great West, published by W.W. Norton & Co. in 1991. This chart shows decreasing travel times between 1830 and 1857 from New York City

to points West. These changes were almost entirely due to the expanding American railway system1.

斯坦福大学的空间历史项目是历史学“空间转向”这一更大潮流的一部分。这是一次谦逊、朴实（同时也昂贵）的尝试，试图以不同方式开展历史研究。我想强调“谦逊”和“朴实”。历史学中许多所谓的“转向”都强调革命性的意图；我认为我们所做的确实不同，但我们并没有宣告人们所熟知的历史已经终结，也没有宣告文本或叙事将要消亡。历史学家仍会写书，仍会讲故事。不过，空间历史项目在五个方面超出了通常的历史实践。第一，我们的项目是协作性的。网站访客看到的许多成果，都是历史学家、研究生和本科生、地理学家、GIS 与可视化专家、数据库架构师以及计算机科学家共同完成的。参与空间历史项目的学者当然可以独自写书，但他们无法独自完成自己所希望规模的空间历史项目：我们缺少知识、技艺，最终也缺少时间。第二，虽然许多展示涉及语言和文本，我们的主要焦点是可视化；这里的“可视化”不仅指地图、图表或图片。第三，这些可视化在很大程度上依赖数字历史。所谓数字历史，我指的就是使用计算机。数字历史使我们能够利用那些若没有计算机就过于晦涩或难以处理的证据和数据库。档案中所有历史学家都会遇到这类材料；我们看过之后往往把它们放在一边。对我而言，铁路货运费率表就是典型例子。第四，这些项目是开放式的：一切东西——工具和数据——都成为学术公地的一部分，可以不断增添、删减、重做和重新组合。最后，也是我们偏离专业规范最关键的一点，是在概念上把空间置于中心。历史学家按定义关注时间。对于试图解释随时间发生的变化的学科而言，年代学始终处于核心位置，但这也使历史学家受到地理学家的指责：他们写历史，仿佛历史发生在针尖上。这一指责并不正确，但有时又令人不安地接近事实。过去半个多世纪的许多经典历史著作——从费尔南·布罗代尔的《地中海世界》到威廉·克罗农的《自然的都市》——都具有空间性，因为变化中的空间关系最能解释随时间变化的格局。威廉·克罗农重新刊印查尔斯·保林绘制的、显示纽约市到美国各地旅行时间变化的旧图表时，就以图形方式表现了时间与空间的相互关系。这是我所谓空间历史的早期例子。图 1 | 查尔斯·保林（1932）绘制的纽约市旅行费率（1830 年与 1857 年）。图表出自威廉·克罗农《自然的都市：芝加哥与大西部》（W.W. Norton & Co., 1991）。它显示 1830 至 1857 年间纽约市到西部各地的旅行时间缩短，这些变化几乎完全源于美国铁路系统的扩张。

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chart from Charles Paullin showing changing times of travel from New York City to various parts of the United States, he gives a graphic representation of the interrelation of time and space. This is an early example of what I am calling spatial history. The problem with citing exceptional spatial histories, however, is that these histories are exceptional. Historians still routinely write about political change, social change, class relations, gender relations, cultural change as if the spatial dimensions of these issue matter little if at all.

What is surprising about this is that in terms of historical theory space has loomed large for a considerable time now. Henri Lefebvre The Production of Space introduced a generation of historians to the idea that space is neither simply natural geography nor an empty container filled by history. It is rather something that human beings produce over time. Spatial relations shift and change. Space is itself historical. Lefebvre, who was a philosopher and not a geographer, organized his own work around three forms of space that he called spatial practice, representations of space, and representational space. Lefebvre, of course, was French and what is often clear enough in French becomes fashionably murky in English. It need not be. Spatial practice is, for example, on one scale, our movement within our homes –from bedroom to kitchen to bathroom to living room. On another scale, it is our movement from home to work along an infrastructure of sidewalks, roads and trains. A further increase in scale creates our long distance movements through airports and along air routes. Spatial practice involves the segregation of certain kinds of constructed spaces and their linkages through human movement.2

Second, there are representations of space. These are the documents of architects, city planners, politicians, some artists, surveyors and bureaucrats.

Figure 2 | Aaron Koblin's Flight Patterns. This still shows some of the flight path patterns of over 200,000 separate flights across the United States in a twenty-four hours period.3 The data for this visualization is from the U.S. Federal Aviation Administration. You can see the entire animation at: www.aaronkoblin.com

Figure 3 | Public Land Survey System. The Public Land Survey System (PLSS), as instituted by the Land Ordinance of 1785, organizes and divides American public lands into a rectangular system of 6-mile by 6-mile township/range squares which are further subdivided into 1-mile by 1-mile sections, numbered one through thirty-six. The illustration above uses Richland County, Wisconsin as an example.4

Figure 4 | Evidence of the Public Land Survey System. Visible evidence of the Public Land Survey System (PLSS) across the United States can be seen above in the aerial landscape photos by Alex MacLean. Above image: "Railroads Across Survey Landscape: Castleton, South Dakota."5

They are not separate from spatial practice because in large measure they are what guide the human labor that creates kitchens, bathrooms, living rooms, roads, train stations, airports, air traffic control and entire landscapes.

Spatial representation is an attempt to conceive in order to shape what is lived and perceived. It is a tremendously powerful and ultimately hopeless set of practices. It is, among other things, the work that James Scott details and condemns in Seeing Like a

然而，援引这些非凡的空间历史存在一个问题：它们之所以非凡，正因为是例外。历史学家仍经常讨论政治变化、社会变化、阶级关系、性别关系和文化变化，仿佛这些问题的空间维度无关紧要，或者几乎无关紧要。令人惊讶的是，在历史理论方面，空间其实已经长期占据重要位置。亨利·列斐伏尔的《空间的生产》使一代历史学家认识到，空间既不只是自然地理，也不是被历史填充的空容器；空间是人类在时间中生产出来的东西。空间关系会移动和变化，空间本身就是历史性的。列斐伏尔是哲学家而非地理学家，他围绕三种空间组织自己的论述：空间实践、空间表征和表征性空间。列斐伏尔是法国人，法语中往往足够清楚的表达，在英语里却时髦地变得晦涩，其实没有必要如此。以一个尺度为例，空间实践是我们在家中的移动——从卧室到厨房、浴室再到客厅；在另一个尺度上，它是我们沿着人行道、道路和铁路，从家到工作地点的移动；尺度再扩大，则是我们通过机场、沿着航线进行的长距离移动。空间实践涉及对不同类型建成空间的分隔，以及通过人的移动把它们连接起来。第二种是空间的表征，即建筑师、城市规划者、政治家、某些艺术家、测量员和官僚所制作的文件。它们并不与空间实践分离，因为在很大程度上，正是这些文件指导着创造厨房、浴室、客厅、道路、火车站、机场、空中交通管制系统乃至整个景观的人类劳动。空间表征试图先进行构想，以塑造人们生活和感知的空间。这是一组极其强大、但最终又注定徒劳的实践。它既是詹姆斯·斯科特在《像国家一样看》一书中详细描述并批判的工作，也是我们许多人在需要分区、下水道、监管、国家公园或荒野地区时所赞赏或追求的工作。图 2｜亚伦·科布林《航班模式》。画面显示二十四小时内美国上空二十多万架独立航班的部分航线模式。可视化数据来自美国联邦航空管理局。完整动画见 www.aaronkoblin.com。图 3｜公共土地测量系统。该系统依据 1785 年《土地条例》建立，把美国公共土地组织并划分为 6 英里 × 6 英里的矩形乡镇／区段方格，再细分为 1 英里 × 1 英里的小区，编号从 1 到
36。上图以威斯康星州里奇兰县为例。图 4｜公共土地测量系统的证据。亚历克斯·麦克莱恩的航空景观照片展示了美国各地公共土地测量系统的可见痕迹。上图：“测量景观中的铁路：南达科他州卡斯尔顿。”

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Figure 5 | Chapel Interior. Churches are spaces redolent with symbolic associations. Religious spaces - like libraries, art galleries, sports arenas and graveyards - are more than locations or buildings. They represent a particular lived experience.7

State, but it is also what many of us praise or seek when we want zoning, or sewers, or regulation, or national parks, or wilderness area.6 These are all representations that we hope will turn into what is perceived and lived.

Finally, there is representational space. This is space as lived and experienced through a set of symbolic associations. It is overlays physical space and makes symbolic use of its objects. It is what marks a church or mosque or synagogue; it is what religious people feel in a sacred space; it is a room in a library or a university building; it is an art gallery. Lefebvre's triad are mutually constitutive; they are not separate categories in an abstract model. Human beings, who create all three, can, but do not always, move seamlessly between them. Lefebvre's triad does not always, or even usually, add up to a seamless or congruent whole. His space, as he admits, is full cracks and fissures.8

Historians' embrace of Lefebvre has been partial. They pay, I think, more attention to the language of spatiality than to the spatial experience, but many of the projects on this web site are more interested in spatial experience and spatial practice than language, and this has forced us to ask myself some hard questions. What operationally do we mean by spatial experience and what specifically are we studying? How does spatial experience connect to the production of space? And, finally, how are spatial relations constructed?

My colleagues have different and more sophisticated answers, but for me, all three of these questions end up having a single answer: movement. I don't want to be so simplistic as to say that if space is the question then movement is the answer, but I fear that I am nearly that simple. We produce and reproduce space through our movements and the movements of goods that we ship and information that we exchange. Other species also produce space through their movements. Spatial relations are established through the movement of people, plants, animals, goods, and information. It would be stupid to argue that narratives and maps cannot represent movement. A history of the Oregon trail or the wonderful maps by C. J. Minard of Napoleon's invasion of Russia both represent movement. We use similar devices in the spatial history project. Maps and texts are critical for representations of space, but representations of space cannot be confined to maps for a simple reason: maps and texts ultimately static while movement is dynamic.10

Figure 6 | Napoleon's March by Charles Joseph Minard.
This classic map by Charles Joseph Minard of Napoleon's March shows the losses sustained by the French Army during the 1812 Russian Campaign. The map succeeds in combining five different variables: the size of the army, time, temperature, distance, and geographic

location.9

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最后是表征性空间：通过一组象征性联想而被生活和体验的空间。它覆盖物理空间，并把其中的对象用于象征性表达；它标记着教堂、清真寺或犹太会堂，是宗教人士在神圣空间中的感受，也可以是一座图书馆或大学建筑中的房间，或者一间艺术画廊。列斐伏尔的三元组彼此构成，并不是抽象模型中的相互独立类别。创造这三种空间的人可以、但并不总能在它们之间无缝转换。三元组也不总是、甚至通常不会汇成无缝或一致的整体；正如列斐伏尔承认的那样，他的空间充满裂缝和缝隙。历史学家对列斐伏尔的接受是部分的。我认为，他们更多关注空间性的语言，而非空间经验；但本网站上的许多项目更关心空间经验和空间实践，这迫使我们提出一些棘手问题：在操作层面，我们所说的空间经验究竟是什么意思？具体研究的是什么？空间经验如何与空间生产相联系？最后，空间关系是怎样构建的？我的同事有不同而且更复杂的答案，但对我来说，这三个问题最终都指向同一个答案：移动。我不想把话说得过于简单，好像“空间是问题，移动就是答案”，但恐怕我几乎就是这么简单。我们通过自己的移动，以及运输货物和交换信息的移动来生产并再生产空间。其他物种也通过移动生产空间。人与人、植物、动物、货物和信息的移动建立了空间关系。说叙事和地图不能表现移动当然很愚蠢。俄勒冈小道的历史，或 C. J. 米纳尔关于拿破仑入侵俄国的精彩地图，都表现了移动。空间历史项目也使用类似手段。地图和文本对于空间表征至关重要，但空间表征不能局限于地图，原因很简单：地图和文本最终是静态的，而移动是动态的。图 5 | 小教堂内部。教堂是充满象征联想的空间。宗教空间——如图书馆、艺术画廊、体育场馆和墓地——不仅是地点或建筑物，还代表一种特定的生活经验。图 6 | 查尔斯 · 约瑟夫 · 米纳尔《拿破仑进军图》。这幅经典地图显示 1812 年俄国战役中法国军队遭受的损失，并成功结合了五个变量：军队规模、时间、温度、距离和地理位置。

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Figure 9 | Botanizing California. This visualization, also from Jon Christensen's Critical Habitats project, was created by Aaron Koblin using data from nearly half a million botanical specimen records stored in herbaria throughout California. It raises questions about the patterns of botanical collecting over time, the construction of knowledge about California's environment, and the ways that these historical sources might be used to explore questions about the present and the future as well as the past.
Figure 7 | Georeferenced USGS Quads in ArcGIS (first) and Historic USGS Quads in Google Earth (second). These illustrations demonstrate the process of georeferencing maps 2D to their physical location on the 3D earth. The first image shows a georeferenced historic USGS quad of San Mateo County. The second illustration shows georeferenced USGS quads of Central California superimposed in the correct space on a Google Earth landscape.
This brings me to new technologies, which are by no means uncontroversial. Historians like myself who advocate the use of these technologies can be accused of having deserted the forces of light and embraced the forces of reaction. The great appeal of GIS to historians is its ability to make historical maps commensurate with modern space and mapping conventions. Because of mistakes in projection even maps like the early U.S.G.S. quads have to be georeferenced. We can geo-reference, or correct them so that they correspond to modern projections of the globe. Anyone who has seen the Google Map site and looked at David Rumsey's georeferenced maps knows what I mean.
ArcGIS is a remarkable, if often time consuming and unwieldy, tool, but it allows the orientation and coordination of dissimilar things –an aerial photograph and a map, for example – in terms of a single location. It allows us to merge things created at dramatically different times to create what are in effect new modern images which potentially reveal things about the past that the original artifacts did not. 11 It also allows us to visualize space in ways that go far beyond mapping, as in the airline routes or as in this representation that Jon Christensen put together for Botanizing California.
Figure 8 | Comparing Layers. This example, taken from Jon Chistensen's Critical Habitat project through the Spatial History Lab, shows how easily different data sets can be compared when they concern the same geographic space. In this case, he compares a digital base map with a USGS quad, an aerial photograph and several digitized polygons signifying geologic soil types and plant habitats. Christensen used this data to investigate the influence of land preservation and conservation on the extinction of the Bay Checkerspot Butterfly. You can read more about his project at the Spatial History Project website: spatialhistory.stanford.edu.

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这让我想到新技术，而新技术绝非没有争议。像我这样倡导使用这些技术的历史学家，可能被指责为背弃光明力量、转而拥抱反动力量。GIS 对历史学家的最大吸引力，在于它能使历史地图与现代空间和制图规范相互协调。由于投影误差，即使早期的美国地质调查局四边形地图也必须进行地理配准。我们可以把它们地理配准，或校正它们，使其与现代地球投影相对应。凡看过 Google 地图网站并查看过大卫·拉姆齐地理配准地图的人，都知道我的意思。ArcGIS 是一种非凡但往往耗时且笨重的工具；不过，它允许我们按照单一地点对方向和坐标不同的事物——例如航空照片和地图——进行定位与协调。它使我们能够把在截然不同的时代创作的东西合并起来，形成实际上全新的现代图像，从而揭示原始文物未必能显现的过去信息。它还使我们能够以远超制图的方式将空间可视化，例如航空路线，或乔恩·克里斯滕森为“加州植物学”项目制作的这幅表现。图 7｜在 ArcGIS 中地理配准的 USGS 四边形地图（第一幅）与 Google Earth 中的历史 USGS 四边形地图（第二幅）。这些图示说明如何把二维地图地理配准到三维地球的物理位置。第一幅显示圣马特奥县的一张历史 USGS 四边形地图；第二幅显示叠加在 Google Earth 景观正确位置上的中加州 USGS 四边形地图。图 8｜图层比较。该例取自空间

历史实验室乔恩 · 克里斯滕森的“ 关键栖息地 ”项目，展示了涉及同一地理空间的不同数据集可以多么容易地进行比较。在这里，他把数字底图、USGS 四边形地图、航空照片以及表示地质土壤类型和植物栖息地的多个数字化多边形进行比较。克里斯滕森用这些数据研究土地保护与保育对湾区棋盘蝶灭绝的影响。更多信息见 spatialhistory.stanford.edu。图 9｜加州植物学。这一可视化同样来自乔恩 · 克里斯滕森的“ 关键栖息地 ”项目，由亚伦 · 科布林利用加州各地植物标本馆保存的近五十万条植物标本记录制作。它提出了关于植物采集随时间变化的模式、有关加州环境的知识建构，以及如何利用这些历史资料探索现在和未来（以及过去）的问题。

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Figure 10 | Absolute Space in Google Maps. Whenever we search online for directions from one place to another, we automatically locate ourselves within a fixed, physical environment. This is “absolute space.” Here, space is measured by distance: inches, feet, meters, miles, etc. Our choice of what route to take when traveling between destinations depends on the distances of different possible routes. However, our choice also depends on less tangible factors: how long it might take to arrive at our destination depending on the time of day during which we are traveling, the cost of transportation (are we driving a car, taking a bus, walking?) or the landscapes through which we might need to travel. Distance belongs in absolute space. The later three factors belong to relational space.12

Figure 11 | Sample Layers: GIS and History, Rio de Janeiro 1840s-70s This visualization, taken from Zephyr Frank’s Terrain of History project at the Spatial History Lab, combines different layers of relational space into one geographic region. Using these layers, we can compare the topography of Rio de Janeiro with property values for different parts of the city, with historic streets and geocoded addresses, all superimposed on a historic base map. Comparing different relational spaces allows us to ask more complex questions and hypothesize connections about the geographic space.
(from top to bottom)
Property value contours--surface created using Kriging (statistical interpolation, prediction map)
This layer is built up by geocoding the addresses of urban property values and then creating an interpolated surface
Digital street layer and geocoded addresses (example here is c. 1870 networked individuals)
Georeferenced historical base map (this map was used to trace the filled topographical contours of the hills of Rio)

GIS is itself a representational space, but its users and sometimes its creators seem oblivious of its representational aspects. Everyday users of GIS rely on the technology to understand where they are on a map and find a route to where they want to be on the map. What they depend on is a correspondence between the map and what I will call absolute physical space. By absolute space, I mean space measured by distance: inches, feet, meters, miles, etc. When we use the technology we mark the fixedness of things: streets, buildings, parks and the rest. Thus GIS often ends up emphasizing not the constructed-ness of space but rather its given-ness, which is fine if you are setting out to bomb something or go out to eat, but not so good if you are trying to understand a wider spectrum of human constructions of space over time.

A georeferenced map is a first step, but because it depends on absolute space, it has definite limits for historians. The first is obvious: not all peoples at all times have constructed space in ways that can be easily made commensurate with absolute space. The second limit of absolute space is that even in Western cultures it is not the dominant space of spatial practice. In everyday experience, people talk about space in terms of miles, but they also do it in terms of time and cost. When space or distance is measured by these non-linear measures, I will call it relational space. In contemporary terms, Palo Alto, where Stanford is located, is nearer to San Francisco at 10:00 in the morning than at 7:30 in the morning or at 5:00 in the evening because it takes less time to get there. Depending on the cost of airline tickets Paris or Melbourne or Shanghai is more or less accessible to Los Angeles at some times than others. This is relational space.

One of the nice things about using ArcGIS is that you actually have to create and think about the different kinds of representation

of space that you are using. You have to construct different layers of space and fit them together. We literally produce representational space in the spatial history lab. These, for example, are some of the layers my colleague Zephyr Frank is working with in his reconstruction of nineteenth-century Rio.
In my own work on transcontinental railroads, I have tried to move between absolute and railroad space in visualizations. Railroads were lines along which machines ran stopping at designated points. This is the infrastructure of railroads and it is best rendered as absolute space. If we imagine these lines through absolute space as the hardware of railroads, then the

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图 10｜Google 地图中的绝对空间。每当我们在线搜索从一个地方到另一个地方的路线时，都会自动把自己定位在一个固定的物理环境中，这就是“ 绝对空间 ”。在这里，空间按距离测量：英寸、英尺、米、英里等。我们在目的地之间选择路线，取决于各种可能路线的距离；但选择还取决于不那么有形的因素：抵达所需时间（取决于出行时段）、交通成本（开车、乘公交还是步行），以及途中可能经过的景观。距离属于绝对空间，后三个因素属于关系空间。图 11｜示例图层：GIS 与历史，里约热内卢 1840—1870 年代。该可视化取自空间历史实验室泽菲尔 · 弗兰克的“ 历史地形 ”项目，把一个地理区域内不同的关系空间图层结合起来。我们可以将里约热内卢的地形与城市不同区域的地产价值、历史街道和地理编码地址进行比较，并全部叠加在历史底图上。比较不同的关系空间，可以提出更复杂的问题，并对地理空间中的联系作出假设。自上而下：地产价值等值线——利用克里金法（统计插值、预测图）创建的表面；数字街道图层与地理编码地址（此处为约 1870 年的网络化个人）；经过地理配准的历史底图（用于描绘里约山丘的填充地形等值线）。GIS 本身就是一种表征性空间，但它的用户、有时甚至它的创作者，似乎都没有意识到其表征性质。GIS 的日常用户依靠技术理解自己在地图上的位置，并找到地图上想去的路线。他们依赖的是地图与我所谓“ 绝对物理空间 ”之间的对应关系。绝对空间指按距离测量的空间：英寸、英尺、米、英里等。使用这项技术时，我们标记事物的固定性：街道、建筑、公园及其他事物。因此，GIS 往往强调的不是空间的建构性，而是空间的既定性。若你要轰炸某物或外出就餐，这样做没有问题；但若想理解人类随时间建构空间的更广泛谱系，就不够理想。经过地理配准的地图是第一步，但由于它依赖绝对空间，对历史学家有明确限制。第一，不是所有时代、所有民族都以容易与绝对空间相协调的方式建构空间。第二，即使在西方文化中，绝对空间也不是空间实践的主导形式。在日常经验中，人们用英里谈论空间，也用时间和成本来谈论。当空间或距离用这些非线性尺度衡量时，我称之为关系空间。在当代语境下，斯坦福所在的帕洛阿尔托，上午 10 点比早上 7 点半或下午 5 点更“ 接近 ”旧金山，因为到达那里所需时间更短。根据机票价格，巴黎、墨尔本或上海在某些时段相对于洛杉矶可能更易到达或更难到达，这就是关系空间。使用 ArcGIS 的好处之一，是你必须创建并思考所使用的不同空间表征，构建不同空间图层并把它们拼合起来。我们在空间历史实验室中确实生产表征性空间；例如，这些就是我的同事泽菲尔 · 弗兰克重建十九世纪里约热内卢时使用的一些图层。在我研究横贯大陆铁路时，我尝试在可视化中往返于绝对空间和铁路空间之间。铁路是机器沿其运行、在指定地点停站的线路；这是铁路的基础设施，最适合用绝对空间表现。如果把绝对空间中的这些线路视为铁路的硬件，那么……

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Figure 12 | Extent of Digitized Railroad. This map from Richard White's Shaping the West project shows the extent of the digitized historic railroads and stations, colored according to company ownership. These rail lines were digitized from the oldest available USGS topographic quads, in the attempt to show the original locations of the railroads.
Figure 13 | Southern Pacific Freight Rates. This example freight table from the Southern Pacific railroad lists the special rates of shipping grain, flour, hay, livestock and wool to market along the railroad. Because wheat was such an important product of California in the late 19th century, these special grain rates are particularly significant. Visualizing the relational space created by these grain rates reveals patterns previously hidden in these dry, numerical charts and allows researchers to begin to understand why the farmers of the San Joaquin valley were so angry with the railroads.13
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A forthcoming series of essays edited by Anne Knowles makes this point, see particularly, Ian Gregory, "A Map is Just a Bad Graph."
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Ibid.
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Freight Rates, Southern Pacific Railroad Company, 1869.
Supplementary Information is linked to the online version of the paper at http://www.stanford.edu/group/spatialhistory/cgi-bin/site/pub_toc.php. Reprints and permissions information is also available at this site.
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rates and schedules were the software, and it was the software that constructed relational space. I don't think we can recover the implications of these rates and schedules without using computers

to visualize the data and the implications for the movement of goods and thus the construction of space.
This visualization is difficult for a casual viewer to decipher, and this is necessarily so, because the visualization is itself a research tool. It is a way of analyzing information more than a finished representation of the conclusion. It is useful for trying to understand how changes in the software (rate table) create changes in spatial relations for producer and consumers in nineteenth-century California.
One of the important points that I want to make about visualizations, spatial relations, and spatial history is something that I did not fully understand until I started doing this work and which I have had a hard time communicating fully to my colleagues: visualization and spatial history are not about producing illustrations or maps to communicate things that you have discovered by other means. It is a means of doing research; it generates questions that might otherwise go unasked, it reveals historical relations that might otherwise go unnoticed, and it undermines, or substantiates, stories upon which we build our own versions of the past.

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图 12｜数字化铁路的范围。这张来自理查德 · 怀特 “ 塑造西部 ” 项目的地图，显示经过数字化的历史铁路和车站范围，并按公司所有权着色。这些铁路线路依据最早可获得的 USGS 地形四边形地图数字化，旨在展示铁路最初的位置。图 13｜南太平洋铁路货运费率。该铁路的示例运费表列出了沿铁路把谷物、面粉、干草、牲畜和羊毛运往市场的特殊费率。由于小麦是十九世纪后期加州的重要产品，这些特殊谷物费率尤其关键。将这些谷物费率创造的关系空间可视化，揭示了此前隐藏在枯燥数字表中的模式，并使研究者开始理解圣华金河谷农民为何如此愤怒地反对铁路。如果把绝对空间中的线路看作铁路的硬件，那么费率和时刻表就是软件，而正是软件建构了关系空间。若不借助计算机将数据及其对货物移动和空间建构的影响可视化，我认为我们无法恢复这些费率和时刻表的含义。这幅可视化很难让普通观众看懂，而这也是必然的，因为可视化本身就是研究工具。它更多是分析信息的方式，而不是对结论的成品呈现。它有助于理解软件（费率表）的变化如何改变十九世纪加州生产者和消费者的空间关系。关于可视化、空间关系和空间历史，我想强调一个重要观点：直到开始这项工作后我才完全理解，而且一直很难向同事充分说明——可视化和空间历史并不是制作插图或地图，去传达通过其他手段已经发现的东西；它是一种开展研究的手段，会提出原本可能不会提出的问题，揭示原本可能未被注意的历史关系，并削弱或证实我们用来构建自身过去版本的故事。参考资料（图注与来源）：William Cronon，《自然的都市：芝加哥与大西部》（纽约：W.W. Norton，1991）；Henri Lefebvre，《空间的生产》（伊利诺伊州芝加哥：Blackwell，1991，第 37—41 页）；Aaron Koblin，《航班模式》（2008）；Township Section Guide（rootsweb）；Alex MacLean，航空景观摄影；James C. Scott，《像国家一样看：改善人类状况的某些方案为何失败》（纽黑文：耶鲁大学出版社，1999）；Keble College Oxford 小教堂内部；Charles Joseph Minard，1812—1813 年俄国战役法国军队损失图（1869）；Anne Knowles 编《放置历史：地图、空间数据与 GIS 如何改变历史学术》（ESRI Press，2007）；Google Maps；Southern Pacific Railroad Company，《货运费率》（1869）。